Carlos Hernandez

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Summary

Electrical engineering student with a concentration in Nanosystems pursuing opportunities in semiconductor fabrication and process engineering. Hands-on cleanroom experience with equipment including spin coaters, UV mask aligners, reactive ion etchers, and atomic force microscopes. Passionate about semiconductor manufacturing challenges and motivated to apply and grow technical skills in high-volume fab environments.

Education

The University of Houston — Houston, TX
B.S. Electrical Engineering, Concentration in Nanosystems | GPA: 3.8/4.0 — *Expected Graduation: May 2028*

Projects

Reactive Ion Etching Process Optimization via Machine Learning — June 2026 – Present
University of Houston Nanofabrication Facility | Undergraduate Researcher — *Houston, TX*

- Built a Bayesian optimization model in Python for Reactive Ion Etching (RIE) process characterization of silicon wafer substrates, exploring the effects of radiofrequency bias power, chamber pressure, and SF_6/Ar gas on etch rate and surface morphology
- Independently prepared and etched 10 silicon samples by executing photolithography-based masking, ICP-RIE etching, and post-etch characterization using atomic force microscopy to record etching process response data
- Measured preliminary etch depths between approximately 200-450 [nm], identifying substrate surface roughness and etch profile irregularities as key targets for process improvement

Wafer Failure Prediction Model — April 2026 – May 2026
University of Houston | Personal Project — *Houston, TX*

- Engineered a silicon wafer failure predictor in Python using Random Forest classification and SMOTE oversampling on SECOM manufacturing data of approximately 1,600 wafers, achieving 99% accuracy and 100% recall on wafer failures
- Developed a data preprocessing pipeline to handle 41,000+ missing values and resolve class imbalance (93/7 pass/fail ratio) using SMOTE, improving model reliability on rare failure cases
- Root-caused wafer failures to identify the top 20 critical process sensors using feature importance analysis

6-DOF Robot Arm — Feb 2026 – Present
University of Houston | IEEE & RoboticsUH — *Houston, TX*

- Engineering a 6-degree-of-freedom robotic arm from the ground up integrating Arduino motor control with MG995 and SG90 servo motors to achieve multi-axis precise positional control
- Implementing embedded control logic for coordinated servo motor actuation across all 6 degrees of freedom, focusing on direct joint angle control before implementing inverse kinematics to define current and future states
- Completed design of the arm's base, incorporating an internal Arduino housing solution for a more compact design

Ultrasonic Radar System — Sep 2025 – Nov 2025
University of Houston | Personal Project — *Houston, TX*

- Designed and built an object-detection radar system using an Arduino, HC-SR04 ultrasonic sensor, and micro servo motor to sweep a 180-degree FOV in under 10 seconds while filtering sensor noise
- Developed a real-time visual radar display in C/C++ to plot object distance and angle data with approximately 0.3 [cm] resolution to simulate a traditional radar sweep interface

Experience

Warehouse Package Handler — Oct 2025 – Present
United Parcel Service — *Houston, TX*

- Load and organize 300+ packages daily into delivery trucks under strict time constraints, optimizing package order to align with delivery route while minimizing delays and preventing misloaded packages
- Developed strong time-management and prioritization skills by meeting productivity metrics during high-volume periods without sacrificing accuracy or safety compliance

Academic Tutor — Sep 2024 – Present
Lone Star College (Kingwood Campus) — *Kingwood, TX*

- Campus tutor for the following courses: Algebra, Pre-Calculus, Calculus I–III; University Physics I-II; Circuit Analysis I-II; General Chemistry for Engineers
- Organize end-of-semester campus-wide tutoring events to assist 100+ students prepare for final exams
- Served as TA for a University Physics II course of approximately 25 students, hosting weekly tutoring sessions and providing additional resources for learning and collaboration via the course Discord server

Technical Skills

Equipment: Spin Coater, UV Mask Aligner, Reactive Ion Etcher, AFM, Oscilloscope, Function Generator, Multimeter
Programming & Software: C/C++, Python, PyTorch, Scikit-Learn, Pandas, Fusion360, LTSpice, Microsoft Office
Electronics: Arduino, Raspberry Pi, Servo Motors, Ultrasonic Sensors, Op-Amps